

A study on the feasibility of using decision models to offload reasoning, and so lower inference cost

Reasoning models spend most of their tokens on questions they would have answered correctly anyway. A **decision model** — small, external, returning a calibrated probability instead of text — is asked per question whether the cheap answer is good enough, and reasoning is bought only when it changes the outcome: **3.1× fewer tokens** at **95%** of reasoning accuracy, or **21% fewer tokens** at no measurable accuracy loss.

Hasan Shabani · deepseek/deepseek-v4-flash-0731 · 660 held-out questions (AIME 2025, GPQA, MATH-500, MMLU-Pro, Stress suite)

3.1× fewer tokens than always reasoning 1,433 vs 4,465 per question	95% of reasoning accuracy kept 84.2% vs 88.5%	+4.6 pts above any equal-budget traffic split 95% CI +1.8 to +7.5	21% saved at no measurable accuracy cost -0.9 pts, McNemar p = 0.48
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COST-ACCURACY FRONTIER

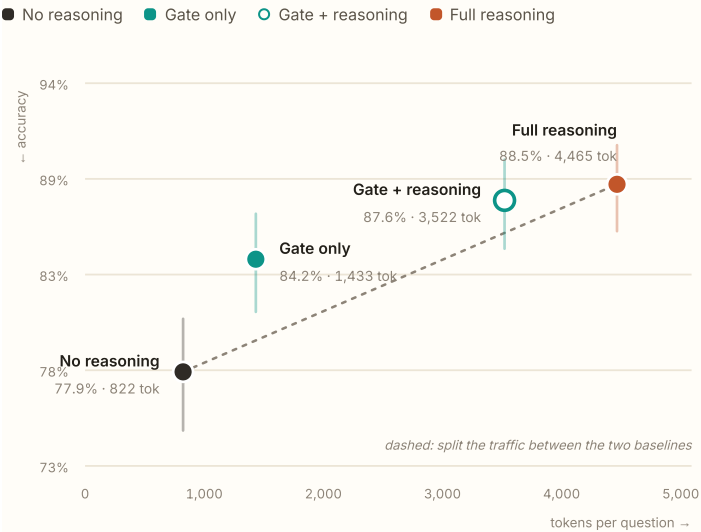
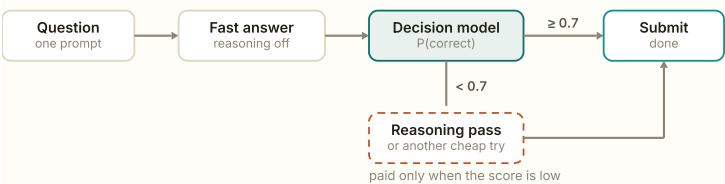


Figure 1. Tokens per question against accuracy; bars are 95% Wilson intervals. The dashed line is the honest baseline — the accuracy bought by routing a share of traffic to reasoning. Both gated policies sit above it; for *gate only* the gap is +4.6 points (95% CI +1.8 to +7.5, paired bootstrap).

MECHANISM



POOLED RESULTS

POLICY	ACCURACY	95% CI	TOKENS	VS REASONING	TOKENS / CORRECT
No reasoning	77.9%	75%–81%	822	-10.6 pts · 0.18×	1,056
Gate only	84.2%	81%–87%	1,433	-4.2 pts · 0.32×	1,701
Gate + reasoning	87.6%	85%–90%	3,522	-0.9 pts · 0.79×	4,021
Full reasoning	88.5%	86%–91%	4,465	reference	5,047

METHOD

- Answer cheaply** — the same model, reasoning disabled, one pass.
- Score it externally** — one yes/no question to a decision model, which returns a probability (1.27 calls per question, \$0.07 per thousand). Self-assessment does not separate right from wrong; an independent scorer does (AUC 0.88 against 0.70).
- Escalate only on doubt** — below 0.7, pay for the reasoning pass (*gate + reasoning*), or retry cheaply and keep the best-scored answer (*gate only*, which never reasons).

FINDINGS

Accuracy is preserved. Gate + reasoning is -0.9 points from always reasoning (22 wins, 28 losses, McNemar p = 0.48) at 21% fewer tokens.

Cheap is not enough on its own. Gate only beats one-shot answering by +6.4 points (p < 0.001) — 1,701 tokens per correct answer against 5,047 for reasoning.

It beats the obvious alternative. The gain over an equal-budget traffic split is +4.6 points, with the confidence interval clear of zero.

EVERY GAP LEFT IS A TRAINING GAP

The model was pretrained and post-trained to reason alone — no way to say "I am unsure", no habit of stopping because something else will check, no experience of being told an answer was rejected. Three measurements point there rather than at the gate: **the judge is not the bottleneck** (AUC 0.88 against 0.70 for self-assessment), **the model cannot act on a rejection** (in the multi-step replay the gate pushed it off a good move 47 times and rescued it 18), and **the right answer is often never proposed** (an audit of the first version's failures found all eleven were cases where no attempt contained it). These are what a gate does *despite* the model. The saving also tracks how much a model overthinks — on a local 35B whose one-shot answers already matched its reasoning it bought no accuracy — and gating each step of an agent episode is the wrong shape (100% solved by reasoning against 53% for the gate on 15 long-horizon tasks).

NEXT STEP

Post-train a model to use the gate, then re-run this benchmark unchanged. The signal already exists: per-question records of the cheap answer, the judge's probability, the outcome and what followed a rejection — enough to teach a model to **defer** (emit its own stop-or-escalate signal) and to **revise** (answer differently when vetoed rather than rephrase). If the gap closes, the ceiling was training.